

[A-38]

SARDAR PATEL UNIVERSITY
BCA (SEMESTER - I)(NC) EXAMINATION (2010 Batch)
Thursday , 16th November ,2017
MATHEMATICS : US01FBCA02
(Mathematics)

Time : 02:00 p.m. to 04:00 p.m.

Maximum Marks : 70

Que.1 Fill in the blanks.

10

- (1) The number of elements in the power set of a set $\{a, b, c, d, e\}$ are
(a) 0 (b) 8 (c) 16 (d) 32
- (2) The set $\{x \in R / 0 < x < 1\}$ is
(a) Finite (b) Infinite (c) Empty (d) none
- (3) If $f(x) = 2x - 3$, then $f^2(2) =$
(a) -1 (b) 2 (c) 0 (d) $4x-6$
- (4) $A = \{1, 3, 5, \dots\}$ is closed under
(a) addition (b) subtraction (c) division (d) multiplication
- (5) $(a^{-1})^{-1} =$
(a) e (b) a^{-1} (c) a (d) a^{-2}
- (6) If group G is commutative then ab ba , $\forall a, b \in G$
(a) < (b) > (c) $\neq$ (d) =
- (7) If $u=(1,2)$ and $v=(2,2)$ then $u + v =$
(a) 7 (b) (3,2) (c) (4,3) (d) (3,4)
- (8) If $u = (a, b)$ and $v = (c, d)$ then $u \cdot v =$
(a) $ab + cd$ (b) $ac+bd$ (c) $ac - bd$, (d) $ad + bc$
- (9) Median of 4, 7, 3, 11, 17, 14, 9, 8 is
(a) 9.5 (b) 9 (c) 8.5 (d) 8
- (10) Mode of 5, 7, 9, 11, 8, 18, 28, 20, 16, 14 is
(a) 11 (b) 12.5 (c) 14 (d) does not exist

Que.2 Answer the following (Any Ten)

20

- (1) Define power set and find the power set of a set $\{a, b, c, d\}$.

(2) Let $f(x) = x^3 + 2x + 1$ and $g(x) = 2x^2 + 1$ find $f \circ g$.

(3) Show that the functions $f: R \rightarrow R$ defined by $f(x) = \frac{3x}{4} - 9$ is one to one.

(4) Define ideal in a ring with example.

(5) Show that right cancellation law holds in a group G .

(6) Find $\frac{(n-1)!}{(n+2)!}$.

(7) Find x, y, z if $(2x, 4, y) = (6, x + z, 2z)$.

(8) Find $(u + v) \cdot w$ and $u \cdot w + v \cdot w$ for $u = (3, -2, 1), v = (5, -3, 4), w = (-1, 6, -7)$.

(9) Find x if $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric.

(10) Find median and mode of 8, 12, 20, 40, 8, 7, 5.

(11) The mean of 200 items was 50. Later on it was found that two items were wrongly read as 92 and 8 instead of 192 and 88. Find out the correct mean.

(12) The intelligence quotients (IQ's) of 10 boys is given by

70, 120, 110, 101, 88, 83, 95, 98, 107, 100

Find the Mean IQ.

Que.3 (a) State and prove Distributive laws.

7

(b) Find the inverse of the real valued function $f(x) = \frac{2x-3}{5x-8}$, $x \neq 8/5$ where it exists.

3

OR

Que.3 (a) If $f(x) = x^2 - 3x + 2$ then find $f(x^3)$, $f(3x - 7)$, $f(x + 5)$

5

(b) By using Mathematical induction Method, prove that $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

5

Que.4 (a) Let Q be the set of all rationals and $*$ defined on Q by $a * b = a + b - ab$.

6

(i) Is $(Q, *)$ a semi group?

(ii) Find the identify element for $(Q, *)$.

(iii) Find $3 * 4$, $2 * (-5)$.

(b) Define identity element of a group and prove it is unique.

4

OR

Que.4 (a) In a ring R , prove that (i) $a \cdot 0 = 0 \cdot a = 0$ (ii) $a(-b) = (-a)b = -ab$.

5

(b) If $G = \{1, 2, 3, 4\}$ is a group under multiplication modulo 5, find multiplication table for it and find $2^{-1}, 3^{-1}, 4^{-1}$.

5

Que.5 (a) If $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 0 & 9 \\ 5 & -7 & 11 \end{bmatrix}$ then find $A + A^T$; $A - A^T$; AA^T 5

(b) If $u = (2, -2, 0, -5, 3)$, $v = (-3, 4, 4, 6, 7)$, $w = (-8, 5, -2, -6, 3)$, $z = (3, -9, 2, -4, 4)$, then find $5u + 3z + 4w - v$, $5u - 4v + 3z$. 5

OR

Que.5 (a) If $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{bmatrix}$, then prove that $(AB)^T = B^T A^T$. 5

(b) Find the inverse of $A = \begin{bmatrix} -1 & 2 & 2 \\ 2 & 3 & 6 \\ 1 & 1 & -7 \end{bmatrix}$, if possible . 5

Que.6 (a) Calculate the mean, median, mode, harmonic mean and geometric mean for the following data. 5

x_i	f_i
10	14
30	23
50	27
70	21
90	15

(b) Eight coins were tossed together and the number of heads (X) resulting was noted. The operation was repeated 256 times and the frequency distribution of the number of heads is given below. Calculate the Median. 5

No.ofheads	Frequency	No.ofhead	Frequency
0	1	5	52
1	9	6	29
2	26	7	7
3	59	8	1
4	72		

OR

Que.6 (a) The following table gives the weights of 31 persons in a sample enquiry. Calculate mean, Harmonic mean, Geometric mean, Median and Mode. 10

Weight	No.ofpersons	Weight	No.ofpersons
130	3	148	5
135	4	149	2
140	6	150	1
145	6	157	1
146	3		

—X—

③

